SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CHENNAI-602105

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SAVEETHA COMPILER ASSESSMENT (LEVEL 1 & VIVA VOCE)

Course Code:	CSA09	**Course Title:**	Programming in Java
CO Assessed:	CO1 – Manipulation (BL4)	**Assessment:**	AT1 – Saveetha Compiler (Level 1 & Viva)
Weightage:	20% of CIA	**Total Marks:**	100 (5 Questions × 20 Marks)
CO1 Statement:	**CO1:** *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3, BL4)*		
Student Name:	L. Sanjay Kumar	**Reg. No.:**	192425226
Section / Batch:	AIML/2024	**Date:**	27-07-2026

Instructions to Students

Answer ALL 5 questions. Total: 100 Marks.
Questions 1–5: Write complete, syntactically correct Java programs on the Saveetha Compiler with proper indentation.
Each question carries 20 Marks.
For program questions: test with at least two sample inputs; ensure output is displayed clearly. Each question carries 5 Marks.
No calculators or electronic devices permitted during the viva session.

Question Type	Description	Questions	Marks
Level 1 – Program Writing	Control Flow, classes and Object programs, Collection Framework on compiler	Q1 – Q4	50
		GRAND TOTAL:	**100 Marks**

SECTION A – LEVEL 1: PROGRAM WRITING QUESTIONS (Q1–Q5)

Q1. (Level 1 – Program Writing) [20 Marks]

Print Numbers from M to N by Skipping K Numbers in the Middle

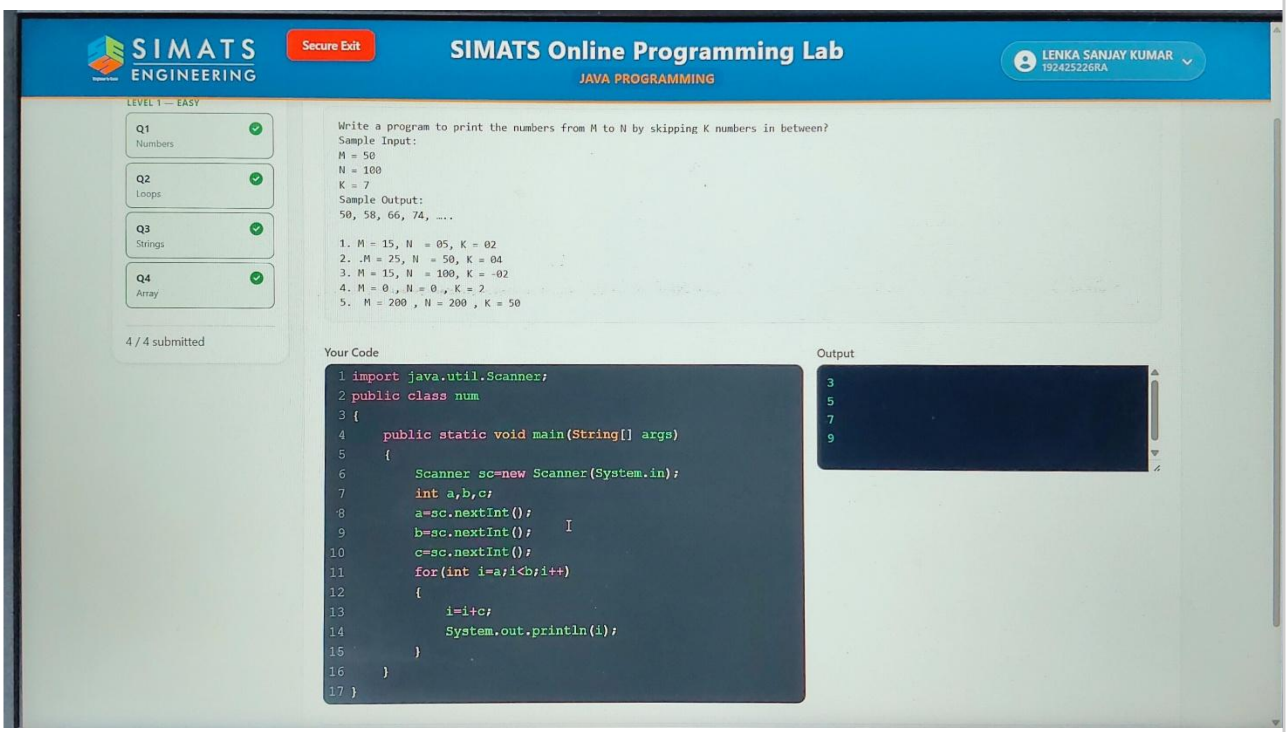
START
Input A
Input B
Input C

mid ← (A + B) / 2
startSkip ← mid - (C / 2)

FOR i ← A TO B
    IF i >= startSkip AND i < startSkip + C
        CONTINUE
    END IF
    Print i
END FOR

STOP

Answer / Program with output:



Q2. (Level 1 – Program Writing) [20 Marks]

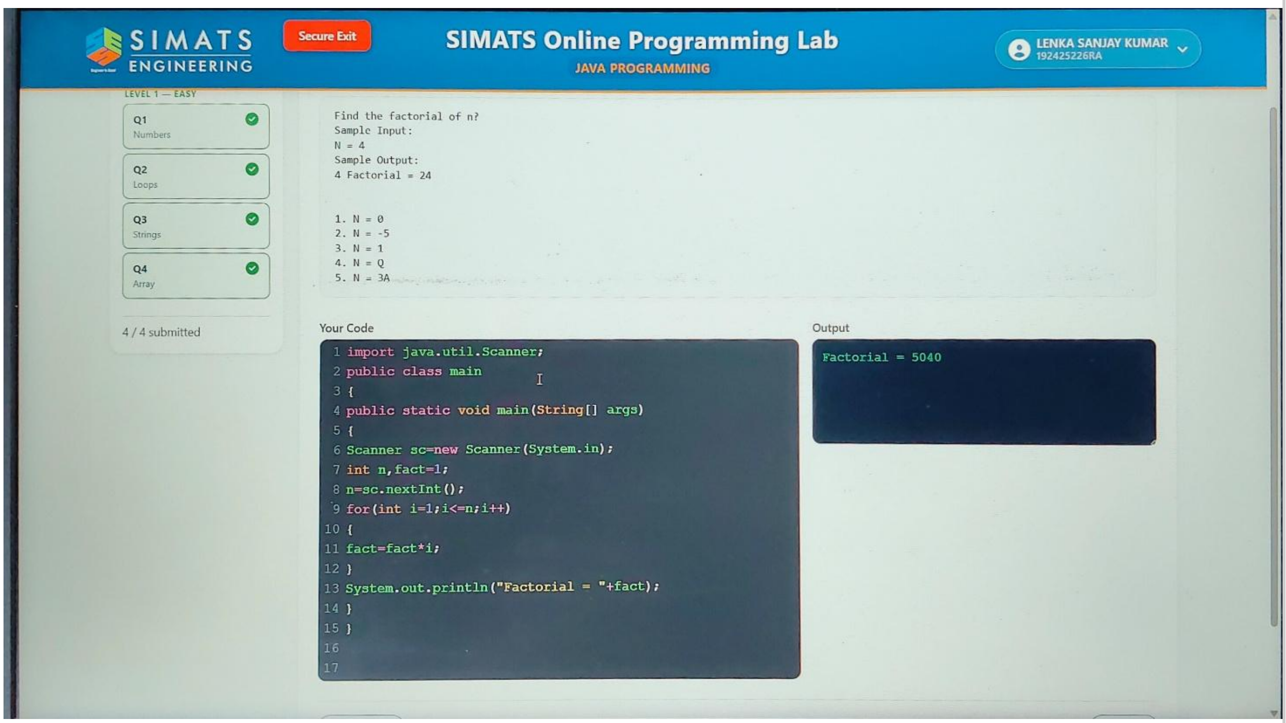
Factorial of a Number

START
Input n
fact ← 1

FOR i ← 1 TO n
    fact ← fact × i
END FOR

Print fact
STOP

Answer / Program with output:

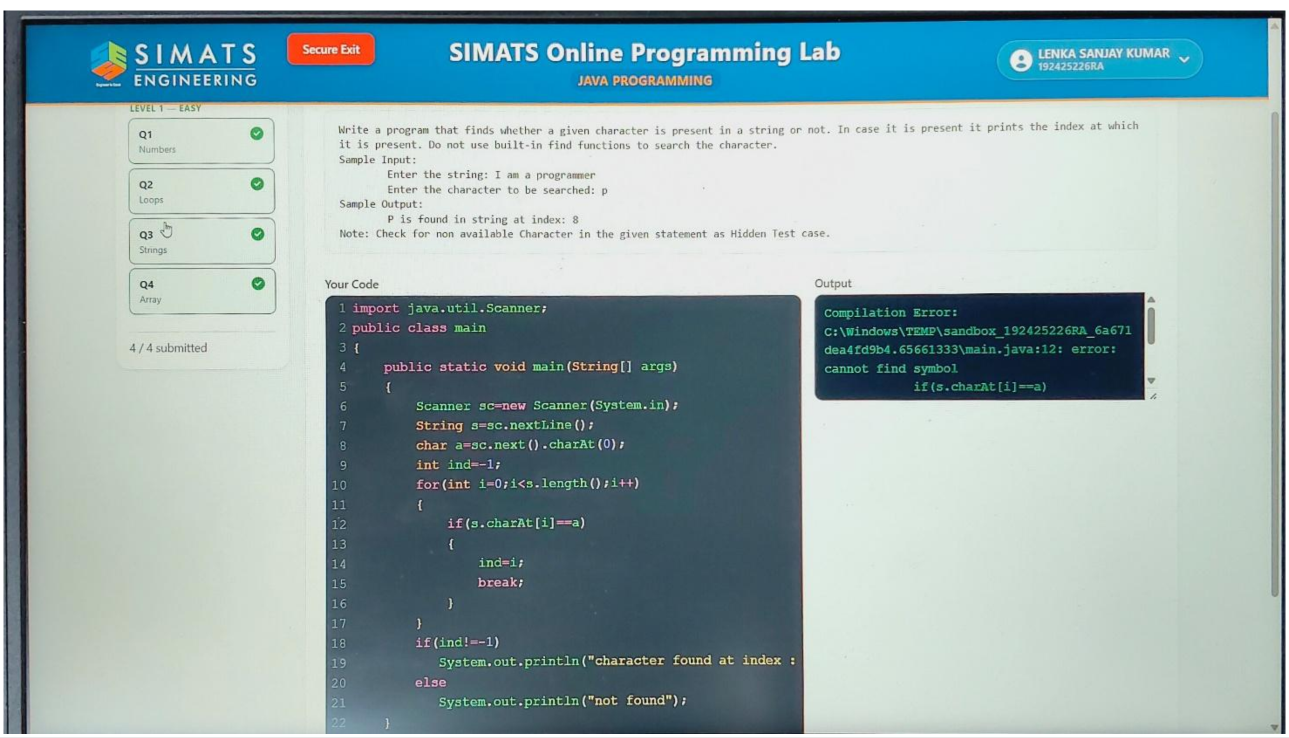


Q3. (Level 1 – Program Writing) [20 Marks]

Find a Character in a String

```
START
Input string
Input character

index ← -1

FOR each character in string
   IF current character = input character
      index ← current position
      EXIT loop
   END IF
END FOR

IF index ≠ -1
   Print "Character found at index", index
ELSE
   Print "Character not found"
END IF

STOP
```

Answer / Program with output:

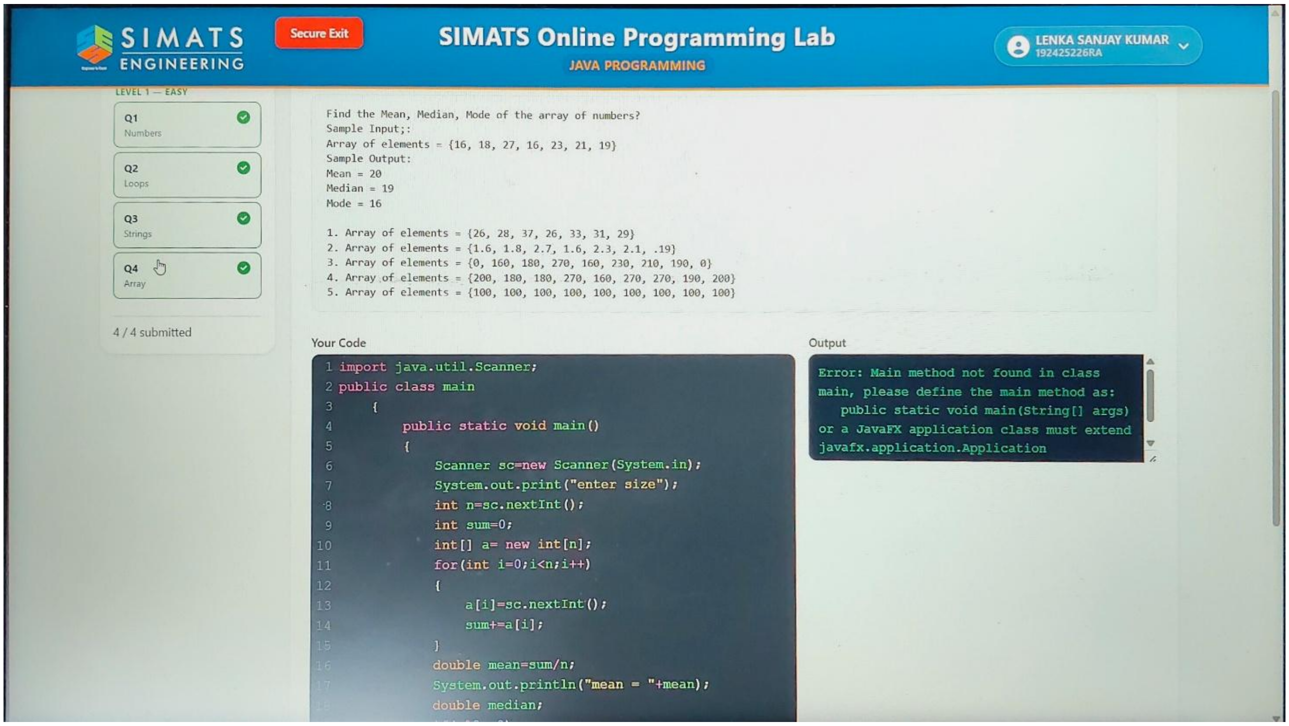


Q4. (Level 1 – Program Writing) **[20 Marks]**

Pseudocode for Mean, Median, and Mode

```
START

Input n
Create array of size n
sum ← 0
FOR i ← 0 TO n-1
   Input array[i]
   sum ← sum + array[i]
END FOR

mean ← sum / n

Sort the array

IF n is even
   median ← (array[n/2 - 1] + array[n/2]) / 2
ELSE
   median ← array[n/2]
END IF

mode ← array[0]
maxCount ← 0

FOR i ← 0 TO n-1
   count ← 1

   FOR j ← i+1 TO n-1
      IF array[i] = array[j]
         count ← count + 1
      END IF
   END FOR

   IF count > maxCount
      maxCount ← count
      mode ← array[i]
   END IF
END FOR

Print "Mean = ", mean
Print "Median = ", median
Print "Mode = ", mode

STOP
```

Answer / Program with output:



RUBRICS FOR EVALUATION

Criteria	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (3–4 Marks)	Level 2 – Developing (2 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Analysis & Understanding				
Logic / Algorithm Design				
Python Syntax & String Operations				
Output Correctness & Testing				
Communication & Explanation (Viva)				

Code of Conduct

I L. Sanjay Kumar/192425226 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: L.S.Kumar